IPCON® PC CF

Carbon Fiber Reinforced Polycarbonate Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.05 mm
Net Filament Weight	1 kg ± 0.5%
Filament Length	335 m ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 90 mm

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.24 g/cm³
Shrinkage Rate*	Specially-defined	0.21% - 0.44%
Surface Hardness	Shore hardness	85D
Saturated Water Absorption Rate	25 °C, 55% RH, room air	0.21%
Melt Index	290 °C, 2.16 kg	14.7 g/10 min
Melting Temperature	DSC, 10 °C/min	226 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	107 °C
Heat Deflection Temperature	ISO 75, 0.45 MPa	113 °C

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527, GB/T 1040	65 MPa
Tensile Strength Z	ISO 527, GB/T 1040	49 MPa
Young's Modulus XY	ISO 527, GB/T 1040	6248 MPa
Young's Modulus Z	ISO 527, GB/T 1040	2362 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	6.5%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	2.1%
Flexural Strength XY	ISO 178, GB/T 9341	116 MPa
Flexural Strength Z	ISO 178, GB/T 9341	73 MPa
Flexural Modulus XY	ISO 178, GB/T 9341	5482 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	2362 MPa
Impact Strength XY	ISO 179, GB/T 1043	23.7 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	16.1 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 290 °C, bed temperature: 100 °C, printing speed: 100 mm/s) and IPCON PC CF filament. The parameters of

specimens of mechanical properties are as following: infill density—100%, infill pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. Ultra-low warping (do not need active chamber heating); ultra-high layer strength; excellent surface quality and accuracy; ultra-low water absorption rate, everlasting strength and stiffness; ultra-high print speed (up to 350 mm/s); upgraded weather resistance.
2. High strength and stiffness prints for water environment applications such as surfboards and water pumps, as well as outdoor applications such as small wind turbines; precise and wear-resistant accessories such as gears.

• Recommended Printing Settings

Drying Settings Before Printing	80 - 100 °C, 8 - 12 h (Blast drying oven)
Printing Temperature and Humidity	≤ 80 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself or support filament for PC
Compatible Printer Type	Enclosed-frame (recommended), open-frame,
AMS or CFS	Not Compatible
Compatible Nozzle Material	Hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.6 mm (recommended) / 0.4 mm / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Apply liquid glue or PVP solid glue stick if the first layer bonding is not firm enough
Nozzle Temperature	280 - 300 °C
Bed Temperature	100 - 120 °C
Fan Speed	0% - 40%
Max Print Speed	350 mm/s

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